A literate Clean specimen

LLM-assisted specimen; see PROVENANCE.md.

1 Determinants of matrix products

This short paper is authored in Clean. Its prose and theorem statements share a source, while the mathematical statements also translate to Lean.

The determinant converts matrix multiplication into scalar multiplication. The explanation is published prose; it introduces no hidden assumptions.

Result 1 (determinant product). [Mathlib]
Multiplicativity for complex square matrices.

Let n be a natural number. Let A and B be $n \times n$ matrices over $\mathbb{C}$. Then

$$\det(A\,B) = \det(A)\,\det(B).$$

1.1 Reversing the order

Since scalar multiplication commutes, the two products AB and BA have the same determinant.

Result 2 (determinant product swap). [Mathlib]
Let n be a natural number. Let A and B be $n \times n$ matrices over $\mathbb{C}$. Then

$$\det(A\,B) = \det(B\,A).$$

The statements above are formal mathematical propositions. These explanatory paragraphs are informal and do not constitute machine-checked proofs.